KHAN INSTITUTE OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY (KICSIT)
DEPARTMENT OF COMPUTER SCIENCE

COMPUTER VISION

PROJECT ASSIGNMENT#2

FIRE AND SMOKE DETECTION USING MOBILENETV2

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1. Introduction

This project focuses on developing a computer vision system to detect fire and smoke using deep learning. Detecting fire and smoke is important for early warning systems and safety applications. A pretrained MobileNetV2 model is used to classify images into fire and smoke categories. The project includes data preprocessing, model training, evaluation, and analysis.

2. Data Preparation

The dataset was manually organized into two classes: fire and smoke.

The dataset was split into:
- **Training:** 70%
- **Validation:** 15%
- **Testing:** 15%

Preprocessing steps included resizing images to 224x224 and normalizing pixel values.
Data augmentation techniques such as rotation, flipping, and zooming were applied to improve model generalization.

3. Model Implementation

A pretrained MobileNetV2 model was used for this task.

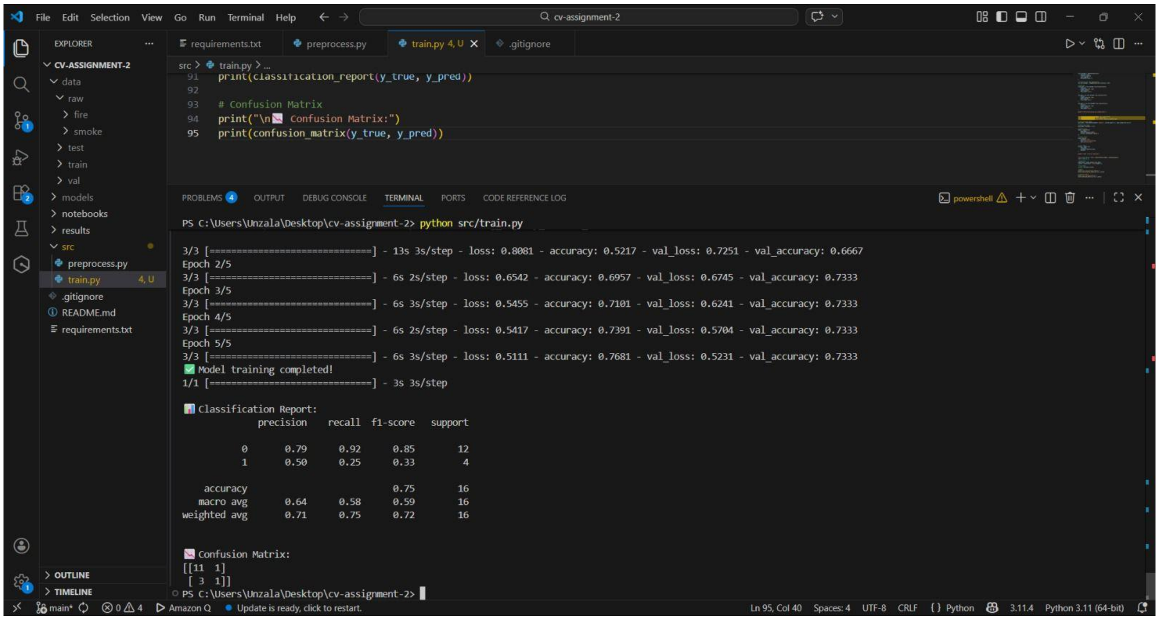
Transfer learning was applied by freezing the base layers and adding a custom classification layer.

Hyperparameters:
- **Learning rate:** 0.001
- **Batch size:** 32
- **Epochs:** 5

The model was trained using the Adam optimizer and binary cross-entropy loss function.

4. Results & Evaluation

The model achieved an accuracy of approximately 75%.
A confusion matrix and classification report were used to evaluate performance.
The model performs well in detecting fire but struggles with smoke detection.



# Confusion Matrix
print("\n Confusion Matrix:")
print(confusion_matrix(y_true, y_pred))
PS C:\Users\Unzala\Desktop\cv-assignment-2> python src/train.py
3/3 [==============================] - 13s 3s/step - loss: 0.8081 - accuracy: 0.5217 - val_loss: 0.7251 - val_accuracy: 0.6667
Epoch 2/5
3/3 [==============================] - 6s 2s/step - loss: 0.6542 - accuracy: 0.6957 - val_loss: 0.6745 - val_accuracy: 0.7333
Epoch 3/5
3/3 [==============================] - 6s 3s/step - loss: 0.5455 - accuracy: 0.7101 - val_loss: 0.6241 - val_accuracy: 0.7333
Epoch 4/5
3/3 [==============================] - 6s 2s/step - loss: 0.5417 - accuracy: 0.7391 - val_loss: 0.5704 - val_accuracy: 0.7333
Epoch 5/5
3/3 [==============================] - 6s 3s/step - loss: 0.5111 - accuracy: 0.7681 - val_loss: 0.5231 - val_accuracy: 0.7333
Model training completed!
1/1 [==============================] - 3s 3s/step
Classification Report:
precision recall f1-score support
0 0.79 0.92 0.85 12
1 0.50 0.25 0.33 4
accuracy 0.75 16
macro avg 0.64 0.58 0.59 16
weighted avg 0.71 0.75 0.72 16
Confusion Matrix:
[[11 1]
[3 1]]
PS C:\Users\Unzala\Desktop\cv-assignment-2>

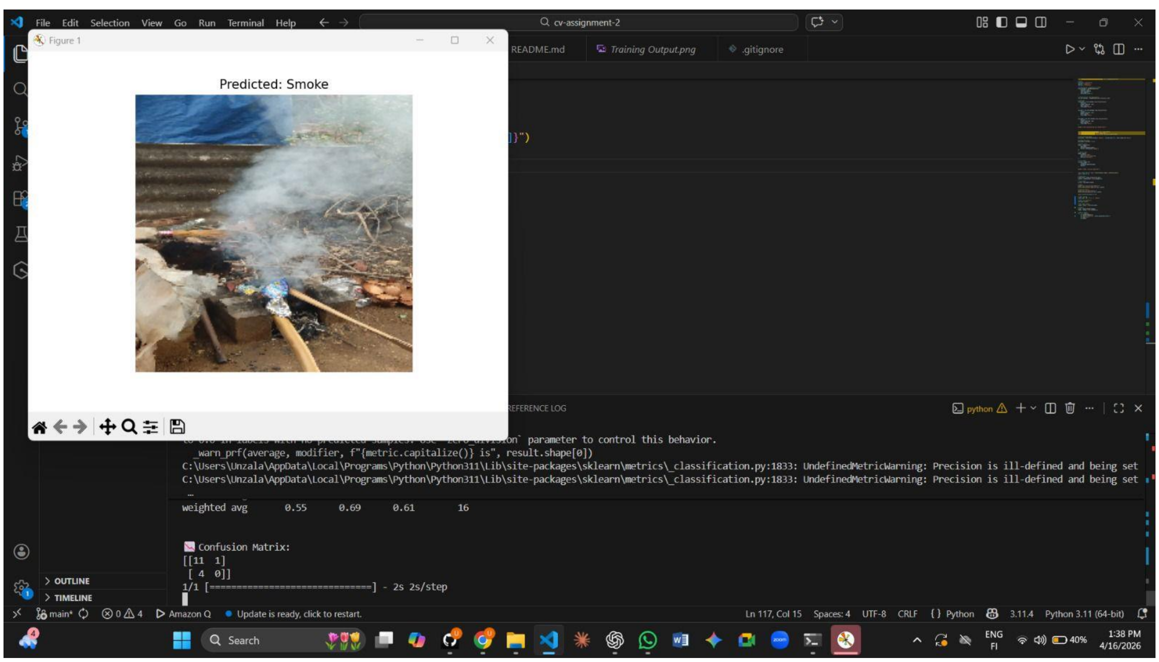
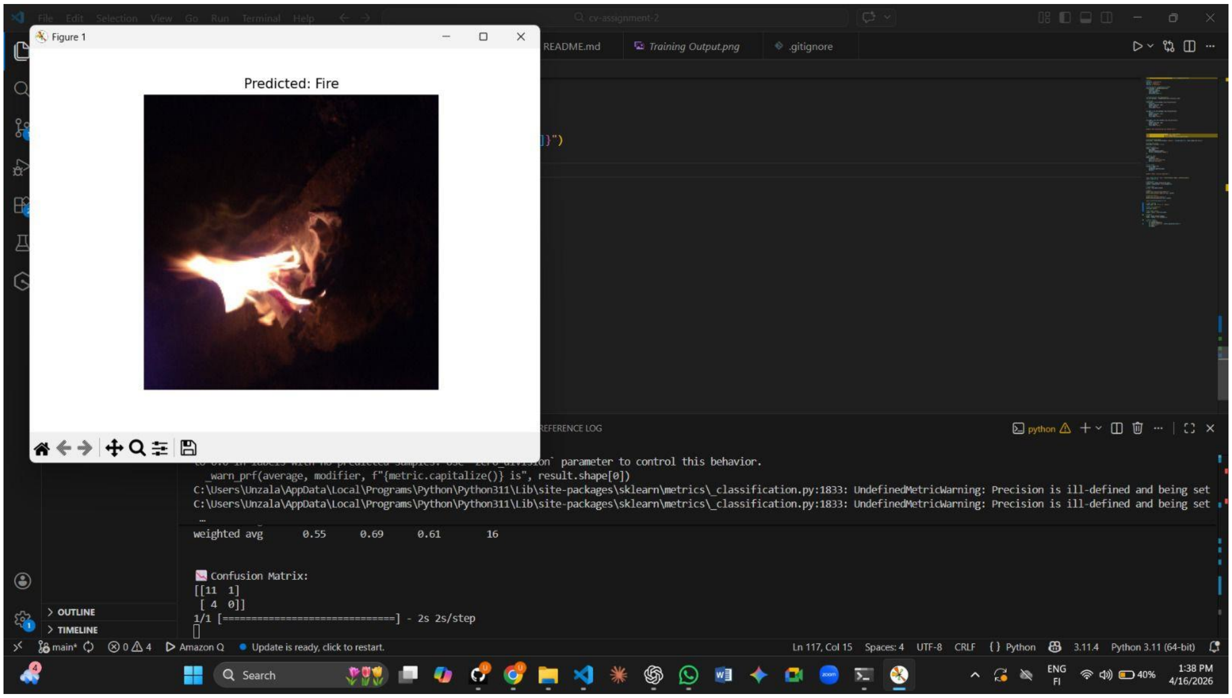


Figure 1
Predicted: Smoke
parameter to control this behavior.
C:\Users\Unzala\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\metrics_classification.py:1833: UndefinedMetricWarning: Precision is ill-defined and being set
C:\Users\Unzala\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\metrics_classification.py:1833: UndefinedMetricWarning: Precision is ill-defined and being set
weighted avg 0.55 0.69 0.61 16
Confusion Matrix:
[[11 1]
[4 0]]
1/1 [==============================] - 2s 2s/step



5. Failure Cases

1. Smoke classified as fire due to similar texture
2. Small fire regions not detected
3. Low light conditions affect detection accuracy
4. Fog or dust misclassified as smoke
5. Blurry images reduce model performance

6. Analysis

Strengths:

- High accuracy for fire detection
- Efficient model using transfer learning

Weaknesses:

- Low performance on smoke detection

Limitations:

- Small dataset
- Class imbalance
- Limited real-world diversity

7. Conclusion

The project successfully demonstrates a working fire and smoke detection system using deep learning. While the model performs well for fire detection, improvements are needed for smoke detection.